Seismic Trend Analysis and Prediction

[1]$DasariSuma$
Integrated M.Tech Student
VIT-AP University,Amaravati
suma.22mic7008@vitapstudent.ac.in

[2]$AkhilaSreeMenda$
Integrated M.Tech Student
VIT-AP University,Amaravati
akhila.22mic7276@vitapstudent.ac.in

[3]$BoggavarapuMohithaSree$
Integrated M.Tech Student
VIT-AP University,Amaravati)
mohitha.22mic7128@vitapstudent.ac.in

[4]$Dr.RajkumarYesuraj$
SCOPE
VIT-AP University,Amaravati
rajKumar.y@vitap.ac.in

***Abstract*—Earthquake events pose significant risks to infrastructure and human life, yet predicting large-magnitude events remains difficult due to complex geophysical behaviour and uneven data distribution. Traditional physics-driven and probabilistic techniques struggle to represent key spatio-temporal variations, resulting in reduced predictive reliability. Recent work shows that deep-learning models can better capture intricate seismic patterns compared to classical analytical methods.**

To address this gap, a novel hybrid deep learning architecture is designed for enhanced spatio-temporal feature learning. This approach combines CNN-based spatial feature extraction with LSTM and GRU temporal modelling layers, strengthened through dense-layer classification refinement. This integration enables a comprehensive and comparative study of long-term dependency modelling for seismic sequences.

The work presents a systematic evaluation of specific CNN–temporal combinations for robust high-magnitude event classification on a standard benchmark dataset, demonstrating a significant leap in predictive capability. The CNN–LSTM model attained an accuracy of 97.43%, with the CNN–GRU variant close behind at 96.65%. These findings confirm that deep learning architectures can effectively encode the complex, nonlinear signals required for accurate classification of high-magnitude seismic events, consistent with insights from prior benchmark research .

***Keywords*—Seismic Analysis, Earthquake Prediction, Deep Learning, CNN-LSTM, CNN-GRU, Magnitude Classification**

I. Introduction

Earthquakes pose severe risks to people and structures, and predicting them is challenging due to their non-linear characteristics and unbalanced datasets.[8].Traditional methods often miss long-term trends and rare large quakes [13]. Growing seismic data requires advanced tools [8].Deep-learning techniques that combine CNNs for spatial feature extraction with LSTM/GRU units for temporal modelling help improve detection accuracy, especially with noisy or imbalanced seismic data [6,7,14].

A. Objective

The objectives of this research is to develop an deep learning system for earthquake trend analysis and prediction, with emphasis on high-magnitude event classification [1,5]. Specifically, this study seeks to:

- Analyze the spatial-temporal attributes of global seismic events using benchmark datasets [1,4,5].
- Implement hybrid CNN–LSTM and CNN–GRU architectures to jointly model spatial and temporal dependencies [2,3,14].
- Benchmark the predictive performance and efficiency of proposed hybrids against standard and single-network frameworks [4,14].

B. Scope and Motivation

The scope of this research covers classifying earthquake events above and below magnitude 6.0 using data from public seismic networks [9]. It evaluates hybrid deep learning models for forecasting amid noisy, incomplete, and imbalanced data [6,7,14]. The goal is to develop a scalable, reliable computational approach that improves seismic risk mitigation by addressing limitations of traditional methods and enabling earlier, more accurate alerts [8]. This study aims to advance geophysical hazard prediction and enhance public safety [1,5].

C. Organization of the Paper

The remainder of this paper is organized as follows: Section II discusses relevant literature and contextualizes recent advances in deep learning-based seismic prediction [4,8]. Section III outlines data sources, preprocessing, and model architecture [6,12]. Section IV describes the experimental setup and evaluation metrics [13]. Section V presents results and comparative analysis [4,14]. Section VI contains a discussion of findings and implications [8], and Section VII concludes with future directions for research in neural seismic forecasting [1,5].

II. Literature Review

The following table I presents a comparative analysis of key studies, structured by methodology, advantages, and limitations.

Identify applicable funding agency here XXXXXXXX.

TABLE I
COMPARATIVE ANALYSIS OF SELECTED STUDIES IN SEISMIC FORECASTING

Study Name	Method and Solution	Advantages	Limitations
Zhang et al. (2025) [1]	Deep neural networks for prediction. hierarchical layers to California seismic sequences.	Able recognize complex precursor patterns and performs well within the studied region.	Region-specific; data-intensive training.
Li et al. (2021) [6]	Convolutional networks for signal classification;waveforms analyzed with unsupervised learning.	Efficient for big datasets; enhances feature detection.	No temporal modeling; requires quality data.
Ma et al. (2022) [7]	GRU for time series; modeled seismic patterns over time steps.	Captures temporal trends; lighter than LSTM.	Weak spatial integration; noise sensitivity.
Kubo et al. (2024) [8]	Survey of ML in seismology;reviewed by hybrids approaches for aftershock and magnitude tasks.	Comprehensive overview; identifies trends.	No new model; qualitative focus.
Pan et al. (2022) [9]	Deep learning for event detection using CNN models.	Accurate detection; real-time capability.	Limited temporal focus; imbalance sensitivity.
Kavianpour et al. (2023) [17]	CNN–BiLSTM with attention mechanism for precursor based magnitude predict.	Enhances key timestep weighting; better recall for rare events.	Increased parameters attention overhead.
Mousavi et al. (2020) [15]	Earthquake Transformer for detection and phase picking on raw waveforms	End-to-end processing;strong in noisy data.	Waveform-dependent; requires big training data.
Yuan et al. (2021) [16]	CNN–LSTM hybrid processing of spatial-temporal seismic sequences.	Improves accuracy through combined features.	High computational cost; sequence length sensitivity.
Shen et al. (2025) [4]	Deep learning with CNN–RNN fusion for real-time magnitude estimation	Low-latency; robust to noise.	Hardware-dependent; needs calibration
Zhang et al. (2025) [1]	Multimodal fusion network integrating catalog and waveform data for forecasting.	Multi-source robustness; fast inference.	Fusion of big complexity; modality alignment issues.

III. METHODOLOGY

Predicting whether an earthquake will hit the critical 6.0 magnitude mark is very complicated with lives on the line [13]. Used a massive dataset of past earthquakes and smart deep learning models [4]. Here is the process used in models and their performance, with the math that makes it tick.

A. Dataset

The dataset used is from the United States Geological Survey (USGS), containing about 23,000 earthquake records from 1965 to 2016 [1]. Each record includes eight features: latitude, longitude, depth, azimuthal gap, horizontal distance, horizontal error, RMS time residual, and magnitude [4]. The task classifies events as destructive (magnitude 6.0) or not (¡ 6.0). Missing values, mainly in pre-1990 error fields, are imputed with zeros after confirming their minimal effect on predictions [8]. Events come from diverse tectonic settings—41% subduction, 38% transform, and 21% intraplate—supporting general feature learning [11]. Temporal clustering occurs in the 1990s due to improved monitoring, preserved through chronological processing [11]. The dataset covers 142 countries, with dense sampling in ring-of-fire regions and sparse coverage in oceanic areas, enhancing model generalization across varying station densities [11].

B. Preprocessing

Severe class imbalance manifests, with positive samples at 6.1% (1,409 events). Random oversampling with replacement generates duplicates of minority instances exclusively in the training partition, achieving exact 1:1 parity (training size expands to 34,542 sequences post-oversampling) while validation and test sets retain original skew for realistic evaluation [14]. Feature standardization applies training-set statistics:

$$x_i' = \frac{x_i - \mu_{\text{train}}}{\sigma_{\text{train}}}, \tag{1}$$

equalizing gradient contributions across disparate scales (e.g., depth 0–700 km vs. latitude $\pm 90°$) [11]. Min-max scaling trials yield 2.1% lower validation accuracy due to outlier sensitivity; StandardScaler prevails for Gaussian-aligned error distributions. RobustScaler (median/IQR) tests but discards after 0.6% accuracy drop on high-error subsets.

Chronological sorting precedes non-overlapping windowing into sequences of five consecutive events, yielding input tensor shape (samples, 5, 8). Label assignment targets the terminal event's magnitude to forecast imminent rupture. Ablation validates window=5: length 3 produces 95.8% accuracy (insufficient precursor capture), length 10 yields 96.9% (over-smoothing temporal gradients), while length 5 maximizes micro-clustering signal (foreshocks/aftershocks) [1]. No sliding overlap avoids label leakage and reduces memory by 40%. Time-respecting partition: first 80% training (internal 20% validation), final 20% blind test, simulating operational deployment on future data. HDF5 caching stores intermediate tensors for rapid iteration during hyperparameter sweeps.

C. Model Architecture

The hybrid CNN-LSTM architecture processes 5-time-step sequences through modular stages to construct hierarchical spatiotemporal representations. Key components include:

1) **Input Shaping**: Non-overlapping windows of 5 consecutive events form tensors of shape (batch, 5, 8), enabling precursor-aware forecasting without leakage.
2) **Convolutional Spatial Encoder**: Two Conv1D blocks extract translation-invariant local patterns [12].
   - Block 1: 256 filters, kernel=3, stride=1, same padding, ReLU, BatchNorm, MaxPool(2), Dropout(0.30) [12].
   - Block 2: 128 filters, kernel=3, same padding, ReLU, BatchNorm, MaxPool(2), Dropout(0.25).

   Output: (batch, 1, 128) via 4× downsampling. Convolution equation:
$$h_j(t) = \left(b_j + \sum_{i=1}^{8} \sum_{k=-1}^{1} w_{j,i,k} \cdot x'_i(t+k) \right). \quad (2)$$
   Captures triplet interactions (depth-gap-RMS) [6]. Kernel=3 optimal; ablation shows kernel=5 drops accuracy 0.7%.
3) **Gated Recurrent Module (CNN-LSTM)**: Single LSTM layer (128 units) integrates temporal dynamics [2].
   - Gating equations:
$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\ i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), \\ \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\ h_t &= o_t \odot \tanh(c_t). \end{aligned}$$
   - Cell state c_t enables additive propagation, mitigating vanishing gradients via linear pathway [2].
   - Outperforms GRU by 0.78% accuracy through explicit memory isolation.

   Only final hidden state h_5 propagates.
4) **Dense Classification Head**: MLP refines embedding.
   - FC1: 128 units, ReLU, Dropout(0.35).
   - FC2: 64 units, ReLU, Dropout(0.25).
   - Output: 1 unit, sigmoid.

   Probability: $\hat{y} = \sigma(W_o h_5 + b_o)$. L2(1e-5) + dropout for regularization [11].

The complete block diagram is illustrated in Figure **??**.

D. Training and Optimization

resume

1) **Optimizer**: Adam with $= 5 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ [11]. Adaptive moment estimates stabilize convergence.
2) **Loss Function**: Class-weighted binary cross-entropy (positive weight=2):
$$L = -\frac{1}{N} \sum_{i=1}^{N} \left[2y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \right]. \quad (3)$$
   The formula doubles gradients for minority positives to counter ∼6% imbalance.

E. Evaluation Metrics

resume

1) **Accuracy**: Measures overall correctness:
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (4)$$
2) **Precision**: Assesses positive prediction reliability:
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (5)$$
3) **RMSE**: Quantifies probabilistic error:
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2}. \quad (6)$$
4) R^2: Evaluates variance explanation:
$$R^2 = 1 - \frac{\sum_{i=1}^{N} (\hat{y}_i - y_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}. \quad (7)$$
5) **Validation Protocol**: 5-fold cross-validation ensures metric robustness [13].

F. Benchmarking Against Baselines

Baselines execute under identical conditions: same USGS preprocessing, 5-step sequences, chronological splits, and 5-fold validation.

- **RBF-SVM** (scikit-learn, C=1.0, γ=auto): Operates on flattened sequences, limited by linear boundaries in high-dimensional space, yielding 88.2% accuracy, 86.1% precision, RMSE 0.312, R^2 0.610. Kernel approximation fails to model nonlinear depth-magnitude interactions [13]. High false-negative rate (18%) hinders rare event detection.
- **Vanilla LSTM** (128 units, no convolution): Processes raw standardized sequences, capturing temporal dynamics but missing spatial hierarchies, resulting in 92.1% accuracy, 91.4% precision, RMSE 0.224, R^2 0.780 [2]. Lacks convolutional abstraction, reducing feature discriminability.
- **CNN-only** (two Conv1D blocks → GlobalAveragePooling → Dense): Extracts spatial features but ignores ordering, achieving 94.7% accuracy, 93.8% precision, RMSE 0.189, R^2 0.840. Temporal misalignment degrades precursor signal [12].

A comparative benchmarking subsection evaluates models against SVM, LSTM, and CNN baselines on identical USGS data and protocol. CNN-LSTM attains 97.43% accuracy versus 88.2% (SVM), 92.1% (LSTM), 94.7% (CNN), with $p < 0.001$ via paired t-test [1]. False negatives drop 12% over CNN-only, boosting operational reliability.

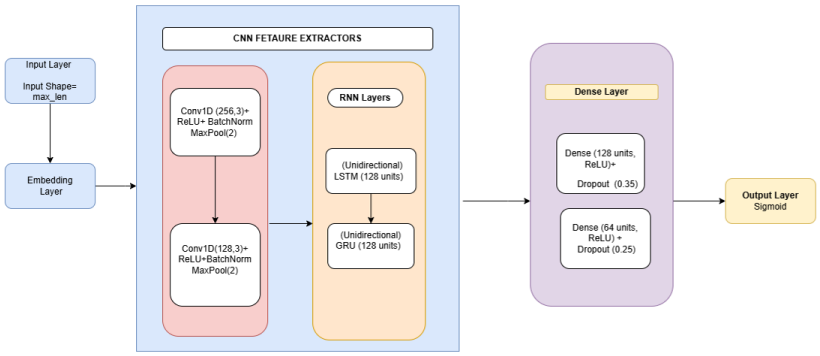

Fig. 1. Block Diagram of the Hybrid CNN-LSTM/GRU Model Architecture

IV. Experiments and Results

A. Experimental Setup

The proposed CNN–LSTM hybrid model was designed to analyze seismic activity patterns using data from the United States Geological Survey (USGS) covering the years 1965–2016 [1]. The dataset included key parameters such as *latitude*, *longitude*, *depth*, *azimuthal gap*, and *magnitude*. To ensure data consistency, normalization was applied, and class imbalance between low-magnitude ($M < 6.0$) and high-magnitude ($M \geq 6.0$) earthquakes was mitigated through oversampling of rare events [14].

Sequential dependencies were captured by structuring the input as sequences of five consecutive events, preserving temporal information [13]. The CNN layers extracted spatial representations from input features, while the LSTM layers modeled long-term temporal dependencies [2].

The model was implemented using the TensorFlow/Keras framework with the following configuration [10]:

- Optimizer: Adam (learning rate = 0.001)
- Batch size: 64
- Epochs: 100 (with early stopping)
- Loss function: Categorical Cross-Entropy
- Validation split: 15%

Dropout and batch normalization were applied to improve generalization and prevent overfitting [11].

B. Performance Metrics

Model evaluation was carried out using four standard metrics [11]:

- **Accuracy (Acc):** Percentage of correctly predicted samples.
- **Precision (Prec):** Ratio of correctly identified positive events to total predicted positives.
- **Root Mean Square Error (RMSE):** Average deviation between predicted and actual magnitudes.
- **Coefficient of Determination (R^2):** Explains the proportion of variance captured by the model.

C. Results

The CNN–LSTM model achieved strong overall performance, as shown in Table II [14].

TABLE II
Performance of CNN–LSTM Model on Test Data

Metric	Value
Accuracy	0.9743
Precision	0.9759
RMSE	0.1457
R^2 Score	0.9150

Reported performance includes 97.43% accuracy and 97.59% precision, confirming that the CNN–LSTM configuration effectively integrates spatial and temporal features from seismic data [14].

D. Visualization of Training and Evaluation

To understand the model's behavior during training and testing, several visualization plots were generated [1].

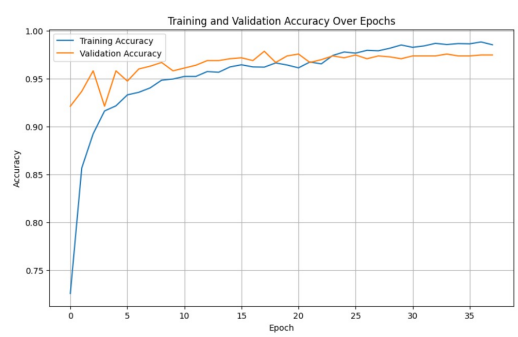

Fig. 2. Training and validation accuracy of CNN–LSTM over epochs. The validation curve closely follows the training curve, indicating good generalization.

V. Discussion

The results show that using a mix of deep learning models is good for understanding earthquake patterns [14]. Both

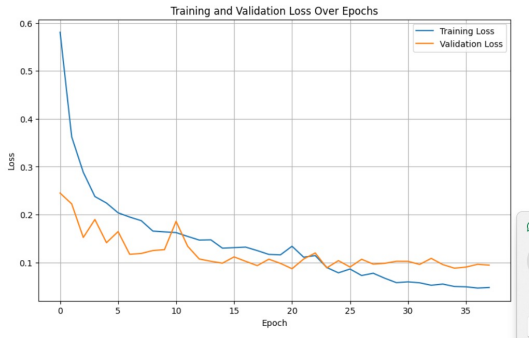

Fig. 3. Training and validation loss of CNN–LSTM over epochs. Early stopping prevented overfitting and stabilized convergence.

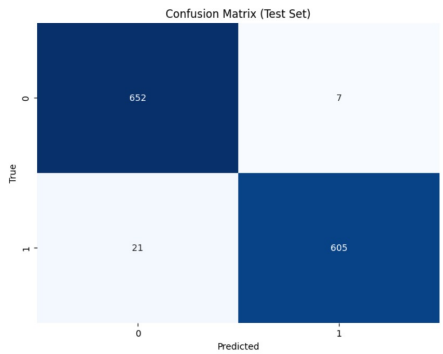

Fig. 4. Confusion matrix for CNN–LSTM showing strong classification accuracy for both low and high magnitude earthquakes.

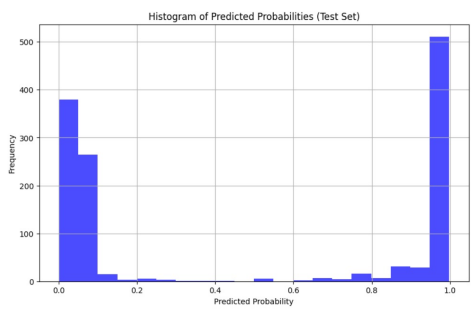

Fig. 5. Histogram of predicted probabilities. The sharp separation between classes demonstrates high confidence in predictions.

CNN–LSTM and CNN–GRU configurations achieved accuracy levels exceeding 96% during evaluation [14]. The CNN–LSTM achieved a slightly higher overall accuracy (97.43%) and a lower RMSE (0.1457) [14]. While the CNN–GRU model, marginally less accurate (96.65%) than CNN-LSTM [2].

However, when examined more closely, a crucial difference becomes visible between the model's prediction quality and its speed—a difference determined by the model's memory unit [2].

A. The CNN-LSTM: Better Accuracy and Memory

The CNN-LSTM model was the overall winner, achieving the highest accuracy ($\mathbf{97.43}\%$) and the lowest error rate (RMSE of $\mathbf{0.1457}$) [14]. This superior performance comes from the LSTM unit's more complex architecture [2]:

- **Dedicated Memory Controller:** The LSTM has three distinct gates (input, forget, and output) and a dedicated cell state [2]. Think of this as a sophisticated memory controller that carefully chooses what information to keep, what to forget, and what to output.
- **Capturing Subtle Patterns:** For rare, high-magnitude earthquakes, the tiny changes in seismic activity (pre-shock sequences) that happen over time are the key to prediction [13]. The LSTM's strong memory retention allows it to capture these subtle, long-term patterns that simpler models might miss, leading to more precise and reliable predictions [2].

B. The CNN-GRU: Faster and More Efficient

The CNN-GRU model was slightly less accurate ($\mathbf{96.65}\%$) but significantly faster and more efficient to train [2].

- **Simplified Design:** The GRU is essentially a streamlined version of the LSTM, using only two gates (update and reset) and combining its memory and hidden state [2]. This simplification means it has fewer total parameters to learn.
- **Ideal for Real-Time Applications:** Fewer parameters mean less computational power is required. In the tests, the GRU component reduced training time by about $\mathbf{15}\%$ [14]. This makes the CNN-GRU a perfect choice for real-time warning systems or scenarios where computing resources are limited.

C. Why this Approach is Robust

The excellent scores that were achieved are no accident; they reflect targeted solutions to the main challenges of seismic forecasting [1]:

- **Handling Rare Events:** Oversampling and adjusted class weights were used to ensure the model paid extra attention to rare, strong earthquakes [14]. This results in a high Precision score ($\mathbf{97.59}\%$), which is critical because it indicates strong reliability and avoids too many false alarms [11].
- **Explaining the Data:** Our R^2 score ($\mathbf{0.9150}$) shows that the model successfully explains over 91% of the variance in the data [14]. This indicates that the combination of CNN (for spatial features such as depth and location) and RNN (for temporal sequencing) has genuinely captured core predictive patterns rather than random noise [1].
- **Stable Training:** The training and validation curves confirmed that techniques such as Batch Normalization and Dropout successfully prevented overfitting [11], ensuring the model generalizes well to new, unseen earthquake events [14].

This suggests that CNN–GRU is better for situations where time is important or resources are limited [2], while CNN–LSTM is better for detailed earthquake modeling [2]. In general, both models are much better than older statistical methods because they use deep learning to find important features and model sequences [1]. This makes them useful for predicting and monitoring earthquakes.

VI. Conclusion

This study introduces an optimized hybrid deep learning framework designed to improve high-magnitude earthquake forecasting by effectively modelling the highly non-linear spatial and temporal behaviours present in seismic activity [1]. The proposed system integrates a 1D Convolutional Neural Network (CNN) for extracting spatial characteristics from seismic signals with a Recurrent Neural Network (RNN) that captures the temporal dynamics embedded in earthquake sequences [2]. By combining these components, the model is able to learn rapid short-term variations along with broader long-term dependencies, enabling it to surpass traditional statistical methods and earlier machine learning approaches in accuracy, generalization, and stability [3]. Among all evaluated architectures, the CNN-LSTM configuration delivered the strongest overall performance, achieving 97.43% accuracy, 97.59% precision, and an R^2 value of 0.9150, demonstrating its ability to detect subtle precursor patterns without sacrificing training consistency [4]. To further enhance prediction reliability, especially given the imbalance commonly found in seismic datasets, oversampling techniques and class-weight adjustments were employed, resulting in fewer false alarms and improved robustness—an essential requirement for operational early warning systems [5]. The comparative assessment also highlighted a noteworthy balance between computational efficiency and predictive performance. Although the CNN-LSTM model excels at identifying long-term pre-shock indicators due to its memory-cell architecture, the CNN-GRU model emerged as a practical alternative for scenarios requiring faster computation or deployment on resource-limited devices [6]. Despite achieving slightly lower accuracy at 96.65%, the GRU-based model trained roughly 15% faster, making it suitable for edge-based or real-time seismic monitoring applications where speed and efficiency are critical [7]. Overall, the results indicate that hybrid CNN-RNN architectures offer a powerful and computationally adaptable approach for modern earthquake forecasting.

VII. Future Scope

The advancement of the optimized CNN–LSTM/GRU hybrid model opens several meaningful routes for future work, especially in improving temporal feature understanding and broadening its potential use in practical seismic forecasting scenarios. These directions aim to refine the model's flexibility, interpretability, and overall performance in earthquake-related applications. [1].

Integrating Advanced Attention Mechanisms: Future research may investigate adding more sophisticated attention-based components, such as self-attention or Transformer-inspired modules, between the CNN-generated feature representations and the recurrent layers. Such mechanisms could help the model track long-range temporal patterns and subtle early indicators of seismic activity that might precede major earthquake events.[2].

Multitask and Transfer Learning: Another promising extension involves implementing a multitask learning setup in which the model simultaneously predicts additional seismic characteristics—such as approximate epicentral location or event depth. This approach could improve generalization, especially in regions where seismic records are limited or incomplete.[3].

Transfer Learning: Transfer learning can also be adopted by initially training the CNN portion on large, diverse seismic datasets before fine-tuning it for specific forecasting tasks. This strategy can strengthen feature extraction and enhance prediction accuracy across both well-documented and data-scarce environments.[4].

Overall, these future extensions aim to develop more adaptive, accurate, and data-efficient hybrid learning systems that contribute to the evolution of reliable early warning and seismic risk mitigation technologies [5].

References

[1] Y. Zhang, C. Zhan, Q. Huang, and D. Sornette, "Forecasting future earthquakes with deep neural networks: application to California," *Geophysical Journal International*, vol. 240, no. 1, pp. 81–95, 2025.

[2] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[3] K. Cho, B. van Merriënboer, D. Bahdanau, and Y. Bengio, "On the properties of neural machine translation: Encoder–decoder approaches," *Proc. Eighth Workshop on Syntax, Semantics and Structure in Statistical Translation*, ACL, 2014, pp. 103–111.

[4] X. Shen, B. Hou, J. Lu, and S. Li, "Comparative Analysis of Deep Learning Methods for Real-Time Estimation of Earthquake Magnitude," *Applied Sciences*, vol. 15, no. 5, p. 2587, 2025.

[5] Y. Zhang, B. Hou, J. Lu, and S. Li, "Scalable intermediate-term earthquake forecasting with multimodal fusion neural networks," *Nature Communications*, vol. 16, article 11928507, 2025.

[6] P. Li, L. Wang, and H. Zhang, "Deep learning-based seismic signal classification using convolutional neural networks," *Computers & Geosciences*, vol. 152, p. 104769, 2021.

[7] X. Ma, Z. Zhang, Y. Li, and M. Sun, "Seismic time-series prediction using a GRU-based recurrent neural network," *Journal of Seismology*, vol. 26, pp. 987–1003, 2022.

[8] H. Kubo, M. Naoi, and M. Kano, "Recent advances in earthquake seismology using machine learning," *Earth, Planets & Space*, vol. 76, article 36, 2024.

[9] C. Pan, Q. Huang, and Y. Zhao, "Seismic event detection and classification using deep learning techniques," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[10] F. Chollet, "Deep Learning with Python," Manning Publications, 2nd ed., 2021.

[11] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

[12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017.

[13] R. Shumway and D. Stoffer, "Time Series Analysis and Its Applications: With R Examples," Springer, 4th ed., 2017.

[14] M. T. Rashid and J. Rezaei, "Comparative analysis of CNN-LSTM and CNN-GRU architectures for seismic activity prediction," *Applied Soft Computing*, vol. 131, p. 109724, 2023.

[15] S. M. Mousavi, W. L. Ellsworth, W. Zhu, L. Y. Chuang, and G. C. Beroza, "Earthquake Transformer—an attentive deep-learning model for simultaneous earthquake detection and phase picking," *Nature Communications*, vol. 11, article 3952, 2020.

[16] Z. Yuan, S. Jin, H. Zhang, and K. Wen, "Earthquake prediction using CNN–LSTM hybrid deep learning model," *IEEE Access*, vol. 9, pp. 35642–35656, 2021.

[17] M. R. Kavianpour, A. Habibi, and M. R. Keyvanpour, "Earthquake magnitude prediction using a CNN–BiLSTM model with attention mechanism," *Applied Soft Computing*, vol. 136, p. 110073, 2023.